

Supplementary Material for Tomorrow Never Knows: A Temporal Knowledge Graph for Music Festival Lineup Forecasting

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A Additional Dataset Details

Figure 1 illustrates additional dataset statistics. Figures 1(a) and (b) depict the number of quadruples over time. Both datasets show steady growth from 1971 to 2019, followed by a sharp drop during the COVID-19 pandemic (2020–2021), which is particularly pronounced in `concertperf`. Quadruple counts recover to near pre-pandemic levels by 2023.

The distribution of relations (Figure 1(c)) in `concert` reveals that concert performances dominate the TKG with 57% of all quadruples (relation *performs concert at*). The high occurrence of the *has type* relation is caused by our temporal encoding strategy: as detailed in Section 4.4, this information is replicated for every year of an artist’s active period, inflating the count of these specific triples.

Regarding the distribution of node types (Figure 1(d)), nodes that represent releases constitute the largest part. This is intuitive: while a single venue hosts a multitude of artists, a single artist may release many albums or EPs, each represented as a distinct node. Figure 1(e) shows the 12 most common release genres: Rock, Electronic, and Pop are the dominant categories, reflecting the mainstream focus of the festivals in the dataset. Figure 1(f) visualizes the geographic distribution of artist areas: although all festivals take place in Germany, the artists are predominantly from Europe and North America, with Germany, the United States, and the United Kingdom contributing the largest shares.

Selected Dataset Facts Beyond aggregate statistics, the dataset enables extracting interesting facts regarding artists and festivals, for example:

- The three largest festival editions in terms of artist line-up size are Reeperbahn Festival 2024 (390 artists), Wacken Open Air 2023 (228 artists), and Wave-Gotik-Treffen 2024 (185 artists).
- The artists who performed at the highest number of festivals in Germany overall are Donots (63 festivals), Emil Bulls (58 festivals), and Itchy (56 festivals).

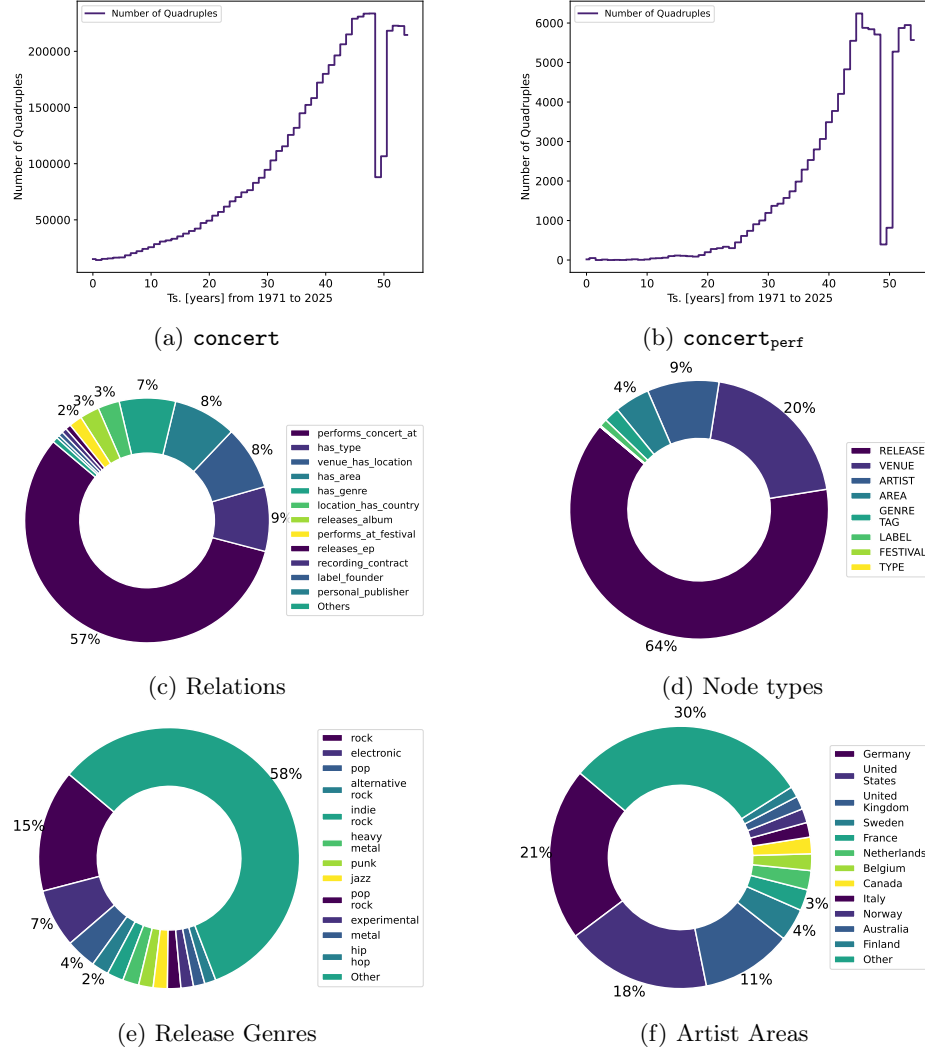


Fig. 1: Top: Number of quadruples over time for **concert** and **concert_{perf}**. Bottom: Pie charts representing distribution of relations, node types, release genres, and artist areas for **concert**.

- The artists who performed at the highest number of festivals in a single year in Germany are Kraftklub (2019; 19 festivals), AnnenMayKantereit (2014; 18 festivals), and Gurr (2016; 17 festivals).
- The artists with the longest festival history in Germany (highest number of festival years) are Guru Guru (31 years; first: 1971, last: 2025), Subway to Sally (30 years; first: 1992, last: 2025), and Project Pitchfork (30 years; first: 1993, last: 2025).
- The venues hosting the highest number of distinct artists are Grote Zaal (2420 artists), Markthalle Hamburg (2310 artists), and Ancienne Belgique (2225 artists).
- The most common origin areas of artists are London (457 artists), Berlin (323 artists), and Los Angeles (267 artists).
- The countries with the highest number of artists (relation *has area*) are Germany (3424 artists), the United States (2877 artists), and the United Kingdom (1799 artists).
- Among artists with available type information (relation *has type*), 73.2% are classified as Group and 26.7% as Person. Additional types (in negligible numbers) include Orchestra, Choir, Other, and Character.

B Experimental Details

For each method, we select the hyperparameters from the ranges specified in the original papers and code repositories. We conduct grid search on the validation set. Table 1 reports the values of selected hyperparameters for each method.

Table 1: Hyperparameter Values selected for each method.

Method	Hyperparameters selected
RE-GCN	n-layers = 2, train-history-len= 4, test-history-len= 4
CEN	start-history-len= 1 dropout= 0.2 n-layers= 2
CognTKE	n-layers = 3, window-size = 15, dropout=0.25 (default hyperparameters, since no ranges or tuning procedure were specified, and results on all datasets in original paper were reported on the same hyperparameter values)
CountTRuCoLa	$\mathcal{P} = 30$, $\mathcal{P}_{f\text{-rules}} = 0$, $\mathcal{C} = \text{True}$, $\mathcal{W} = 20$, $\mathcal{M} = 50$, $\mathcal{Z} = 0.01$, $\mathcal{H} = 10$, $\mathcal{D} = 0.4$, $\mathcal{C}_{Xcount} = 3$; for concert _{with shortcuts} we set $\mathcal{C}_{Xcount} = 10$, due to larger TKG size.
TLogic	num-walks= 10, transition-distr= <i>exp</i> , window= 30, top-k= 10, $\alpha = 1$, $\lambda = 0.01$, rule-length= [1, 2, 3]
Rec.	No Hyperparameters

C Details on LLM Experiments

C.1 LLM Prompts

The prompts used for the LLMs were evaluated and tuned on an iterative approach, generating two prompts for artists and festivals repeatedly¹.

For receiving artists playing at a festival, the LLMs were prompted like this:

You are estimating likely performers for the `\{name\}` `\{year\}` in `\{city\}`, `\{country\}`.

Task:

Generate a ranked list of 210 artists most likely to perform.

Output requirements:

- Provide exactly 210 artist names
 - Rank them from 1 (most likely) to 210 (least likely)
 - Output ONLY the ranked list
 - Format: each line should be "1. Artist Name", "2. Artist Name", etc.
 - Do NOT include explanations, notes, or extra text
- If uncertain, still provide your best estimate based on the criteria above.

For receiving festivals where an artist will play, the LLMs were prompted like this:

You are estimating which festivals in Germany an artist named "`\{name\}`" is most likely to perform at in `\{year\}`.

Task:

Generate a ranked list of 50 German festivals

where "`\{name\}`" is most likely to perform in `\{year\}`.

Output requirements:

- Provide exactly 50 festival names
 - Rank them from 1 (most likely) to 50 (least likely)
 - Output ONLY the ranked list
 - Format: each line should be "1. Festival Name", "2. Festival Name", etc.
 - Do NOT include explanations or any extra text
- If uncertain, still provide your best estimate based on the criteria above.

C.2 LLM Ranking Quality

In the following, we show a ranking produced by Llama 3.3 70B when asking who will perform at the *Keep It True Festival* in 2025. Female artists and

¹ The prompts, as well as input and evaluation code are also available on GitHub: <https://github.com/JuliaGast/festival-lineup-tkg/tree/main/forecasting/llm>.

female-fronted bands are marked in green. Rather than being spread throughout the ranking as knowledge of the festival would suggest, these acts form a contiguous block. Artist gender is not a meaningful criterion for ranking bands at a traditional heavy metal festival, so the observed clustering exemplifies the auto-regressive bias described in Section 5.4: once the model generates one female-fronted act, subsequent token probabilities shift toward similar acts.

- | | |
|----------------------|-----------------------|
| 1. Saxon | 21. Turisas |
| 2. Judas Priest | 22. Sabaton |
| 3. Iron Maiden | 23. Amon Amarth |
| 4. Scorpions | 24. Children of Bodom |
| 5. Helloween | 25. In Flames |
| 6. Blind Guardian | 26. Dark Tranquillity |
| 7. Accept | 27. At the Gates |
| 8. U.D.O. | 28. Arch Enemy |
| 9. Sabaton | 29. Nightwish |
| 10. Powerwolf | 30. Epica |
| 11. Rhapsody of Fire | 31. Within Temptation |
| 12. Hammerfall | 32. Delain |
| 13. Stratovarius | 33. Leaves' Eyes |
| 14. Sonata Arctica | 34. Xandria |
| 15. Dragonforce | 35. Lacuna Coil |
| 16. Alestorm | 36. Evanescence |
| 17. Korpiklaani | 37. Kamelot |
| 18. Eluveitie | 38. Symphony X |
| 19. Ensiferum | 39. Evergrey |
| 20. Finntroll | 40. Threshold |

D Additional Experiments

D.1 Effect of LLM Knowledge Cutoff

Since the knowledge cutoff of the evaluated LLMs falls around late 2023, queries about 2025 festival lineups may fall outside their reliable knowledge window. To assess whether this introduces a systematic bias, we compare model performance across three years, 2023, 2024, and 2025, using a random subset of 50 artists and 50 festivals sampled from quadruples occurring in all three years. To limit expenses, these tests were only conducted on the three open-source models.

Figure 2 (top) reports Mean R-Precision per year and model. CountTRuCoLa consistently outperforms all LLM-based approaches across all years, consistent with the main results in Section 5.3. To analyze year-to-year differences, we compute the relative deviation of each model’s per-year score from its three-year mean, shown in Figure 2 (bottom). The deviations are comparable in magnitude across LLMs and CountTRuCoLa alike: Llama 3.3 70B, Mixtral, and CountTRuCoLa peak in 2024, while Llama 3.1 8B reaches its lowest score in that year. Crucially, no model exhibits systematically stronger performance in 2023

or 2024 relative to 2025, and the LLMs do not show unusually large temporal fluctuations compared to CountTRuCoLa.

We find no evidence that the knowledge cutoff substantially degrades 2025 performance, and therefore proceed with 2025 data in all subsequent experiments.

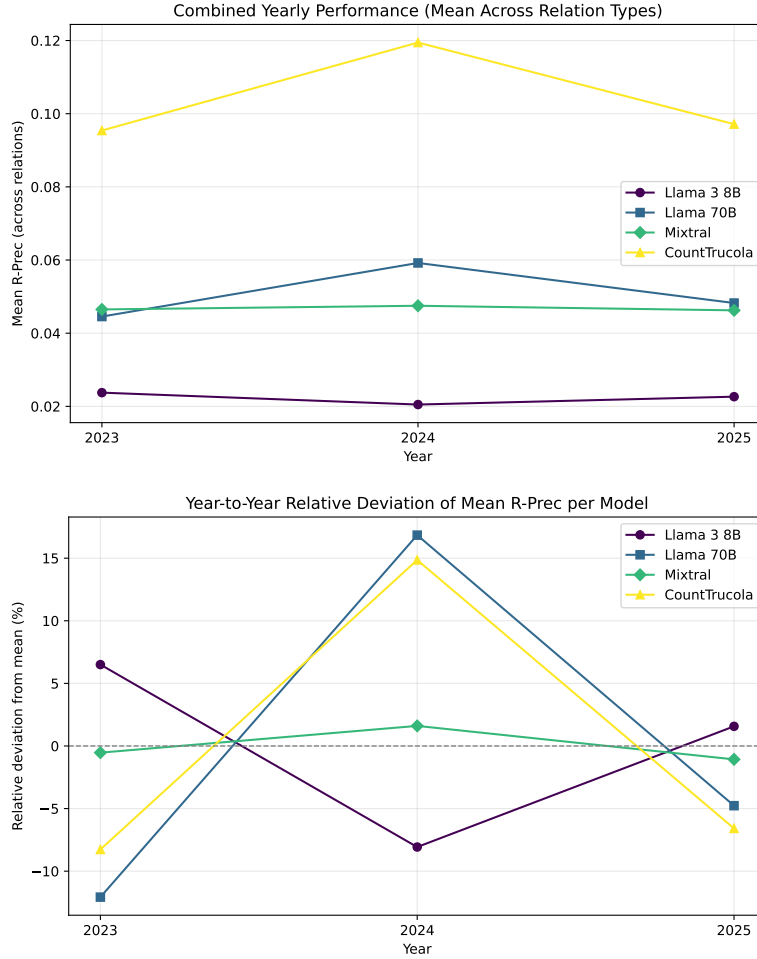


Fig. 2: Per-year performance on Mean R-Precision (top) and relative deviation from each model’s three-year mean (bottom), evaluated on a subset of 50 artists and 50 festivals appearing in all three years.

D.2 Addition of shortcut relations

To address the challenge that multi-hop inference across distant nodes may pose for certain TKG models, we introduce auxiliary shortcut relations under the category *Additionally Added Shortcuts*. These relations do not introduce new information but provide direct edges between artists and attributes (e.g., countries where they perform, genres of their releases) thereby offering an alternative path to information that would otherwise require traversing multiple hops. While this creates redundancy and increases graph size, it may benefit models with limited hop capacity.

We treat shortcut relations as optional input and conduct experiments under two configurations: with and without shortcuts. Table 2 details the schema extensions, and Table 3 reports dataset statistics for `concertshortcuts` alongside the default `concert` and the reduced `concertperform`. While `concert` and `concertshortcuts` share identical node counts, due to the additional relations injected in `concertshortcuts`, the number of quadruples increases. Importantly, the prediction task as well as the ground truth, are consistent across both variant: the *performs at festival* quadruples exist unchanged in all three datasets, ensuring that test-set results are directly comparable.

Table 4 reports results for CountTRuCoLa and TLogic on both dataset variants, `concertshortcuts` and `concert`. The remaining methods could not be evaluated due to out-of-memory errors. No consistent improvement is observed when adding shortcut relations, and we therefore conclude that, at least under the current model capabilities, they do not provide a measurable benefit.

Table 2: Data schema, extended triples, for the temporal KG. Each temporal triple also comes with a timestamp.

	Domain	Relation	Range
Additionally Added Shortcuts		ARTIST releases album of genre	GENRE-TAG
		ARTIST releases ep of genre	GENRE-TAG
		ARTIST performs festival in country	AREA
		ARTIST performs festival in location	AREA
		ARTIST performs concert in country	AREA
		ARTIST performs concert in location	AREA

Table 3: Dataset statistics for all three dataset variants: The number of nodes (#N), relations (#R), train, valid, and test quadruples, the time interval, number of train/valid/test timestamps, Direct Recurrency Degree (DRec), and Recurrency Degree (Rec).

Dataset	#N	#R	#Train	#Valid	#Test	Time Int.	#Tr/Val/Te TS	DRec [%]	Rec [%]
concert	278.911	23	5.068.265	222.507	214.578	1 y.	53/1/1	26.9	44.2
concert_{shortcuts}	278.911	29	8.927.809	423.314	409.911	1 y.	53/1/1	25.0	53.1
concert_{perform}	25.399	1	83.889	5.949	5.570	1 y.	51/1/1	2.9	29.4

Table 4: Test results per direction: predicting artists (*?, performs at festival, F, 2025*), predicting festivals (*A, performs at festival, ?, 2025*), and predicting both directions, i.e. aggregated over all test queries. Best results are **bold**, second best are underlined. Comparing the default **concert** dataset to the extended **concert_{with shortcuts}** dataset version.

	predicting artists					predicting festivals					both directions				
	MRR	H1	H10	P _R	P _{N10}	MRR	H1	H10	P _R	P _{N10}	MRR	H1	H10	P _R	P _{N10}
concert															
CountTRuCoLa	2.4	0.9	4.4	7.7	10.5	23.0	13.5	42.5	16.5	40.4	12.7	7.2	23.5	16.0	<u>38.5</u>
TLogic	0.9	0.3	1.5	3.8	4.3	11.3	6.5	22.9	9.3	21.6	6.0	3.2	12.2	8.7	20.6
concert_{with shortcuts}															
CountTRuCoLa	2.5	1.0	4.5	8.2	10.5	22.5	13.6	40.9	16.7	39.0	12.5	7.3	22.7	16.2	37.3
TLogic	0.9	0.1	1.5	3.1	2.9	11.5	6.6	22.8	9.5	21.4	6.1	3.3	12.3	8.8	20.4